BROKE STREET BETS

Commodity–Stock Links and Market Regimes

Bayesian VECM, directed networks and MS-GARCH

Team members

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Research design and data

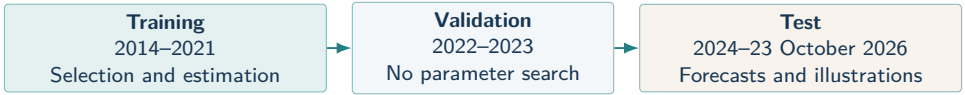
Do commodity and macroeconomic hubs help explain equity dynamics and volatility?

Element	Specification	Role
Equities	500 Indian companies	Stock-price responses
Hubs	80 commodity and contextual series	Agriculture, metals, energy, FX, rates and macro
Potential pairs	40,000 per horizon	Full cross-product of hubs and equities
Observation period	January 2014–23 October 2026	Daily, weekly and monthly analyses
Data availability	37 hubs (daily/weekly); 73 (monthly)	Available histories vary by horizon
Sources	LSEG Workspace; RBI, MoSPI, PIB and price archives	Market and macroeconomic histories



The study evaluates predictive associations; structural causality is not identified.

Sample construction and evaluation design



Choice	Setting	Design rationale
Training sample	Longest joint contiguous block; earliest tie	Avoid interpolation and inconsistent lags
Minimum levels: D / W / M	252 / 52 / 36 observations	Retain a basic estimation history
Difference lags: D / W / M	5 / 2 / 2; HAC bandwidth matches lag	Represent short-run dependence consistently
Pair selection	Engle–Granger raw $p < 0.05$	Broad exploratory relationship screen
Integration and multiplicity	ADF/KPSS; BH/BY, BH $q \leq 0.10$	Report diagnostic context separately
Correction reference family	80,000 directions per horizon/test	Preserve the full universe denominator

Selected samples: 561 daily, 1,247 weekly and 1,943 monthly pairs. The screen was broadened after an initial stricter analysis; findings are exploratory. No validation/test parameter search was performed.

Bayesian error-correction specification

$$\Delta z_t = a\big(s_{t-1} - b x_{t-1} - m\big) + \sum_{j=1}^{L} \Gamma_j \Delta z_{t-j} + e_t, \qquad e_t \sim N\big(0, \operatorname{diag}(\sigma_s^2, \sigma_x^2)\big)\,.$$

Parameter / convention	Specification	Design rationale
$z_t = (s_t, x_t)^\top$	Log stock; log hub when positive in training	Represent proportional changes
Native-level hub scaling	Training mean and population SD only	Normalize native-level variation
Long-run slope b	$N(1, 0.5^2)$	Baseline slope regularization
Equilibrium intercept m	$N(0, 1^2)$	Allow an equilibrium offset
Adjustment vector a	$N(0, 0.1^2)$ componentwise	Shrink toward slow adjustment
Short-run effects Γ_j	$N(0, 0.2^2)$ componentwise	Limit lag-effect dispersion
Noise scales σ_s, σ_x	Half-Normal(0.1); diagonal covariance	Positive error scales
Rank and lag order	Rank one; $L = 5/2/2$ (D/W/M)	Match the bivariate baseline

For non-I(1) pairs, these are conditional ECM fits rather than validated cointegrating systems.

Posterior estimation and quality criteria

Choice	Value	Design rationale
Sampler	NUTS; 2 chains	Assess between-chain agreement
Retained draws	2,000 per chain	Estimate posterior summaries and intervals
Warm-up; acceptance target	1,000; 0.90	Adapt proposals conservatively
One diagnostic retry	2,000 warm-up; target 0.95	Address difficult posterior geometry
Convergence	$\widehat{R} \leq 1.01$	Require close between-chain agreement
Bulk and tail ESS	≥ 400 for both	Require usable effective sample sizes
Energy diagnostic	BFMI ≥ 0.30 in each chain	Check exploration of energy levels
Divergences	Zero	Reject unresolved numerical trajectories
Intervals; predictive draws	95% HDI; 500 predictive draws	Summarize uncertainty and training fit
Likelihood verification	100 prior points; QR relative SSE error $\leq 10^{-10}$	Check exact likelihood computation

Exploratory equilibrium support additionally requires I(1) diagnostics, no recorded source break, a negative adjustment HDI and stable posterior-mean companion dynamics. Half-life is reported only for a stable, positive, monotone impulse response.

Bayesian and directional test results

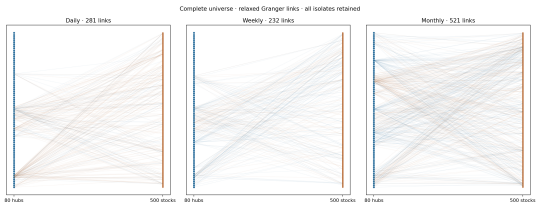
Horizon	Attempted	Sampler pass	Sampler fail	Equilibrium support
Daily	561	264	297	40
Weekly	1,247	716	531	140
Monthly	1,943	1,014	929	66
Total	3,751	1,994	1,757	246

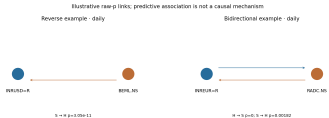
Directional tests: significant directions, both directions counted

Horizon	Granger $p < .05$	Granger BH	ECM short run	ECM adjustment	Market control
Daily	281	60	270	678	122
Weekly	232	27	306	1,477	227
Monthly	521	91	432	2,088	414

BH column uses $q \leq 0.10$; other directional columns use raw $p < 0.05$. Corrections are diagnostic after pair selection. Equilibrium support is an additional exploratory classification, not a count of causal relationships.

Directed predictive network

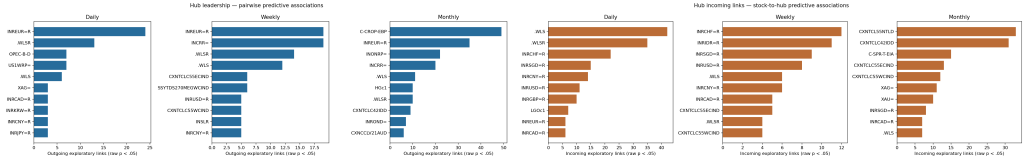




Horizon	Raw-p links	Active nodes	Isolated nodes	Largest component
Daily	281	154	426	154
Weekly	232	149	431	134
Monthly	521	227	353	227

The universe contains 580 nodes. There are 7,502 tested directions across 3,751 pair-frequency models; the table counts only significant raw-p links. Arrows indicate predictive direction.

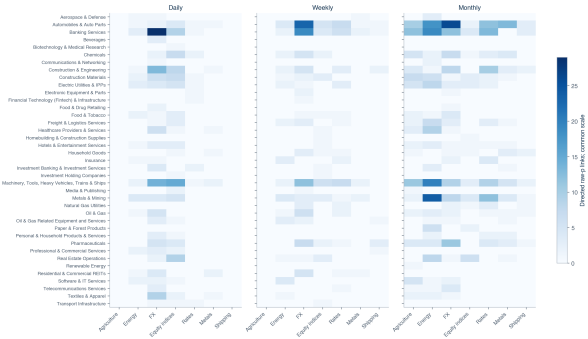
Hub influence and reverse predictability



Horizon	Largest outgoing hub	Links	Largest incoming hub	Links
Daily	INREUR=R	24	.WLS	42
Weekly	INREUR=R	19	INRCHF=R	12
Monthly	C-CROP-EBP	49	CXNTCL55NTLD	33

Outgoing: hub predicts stocks. Incoming: stocks predict the hub. Rankings reflect observed sample coverage and the exploratory pair screen.

Commodity categories and equity sectors



Horizon	Hub category	Equity sector	Directed links
Daily	Foreign exchange	Banking Services	29
Weekly	Foreign exchange	Automobiles & Auto Parts	23
Monthly	Foreign exchange	Automobiles & Auto Parts	26

Common color scale across horizons. Counts represent associations, with coverage and sample-selection effects retained.

Regime definitions and thresholds: 1–5

Dimension	States and classification	Design rationale
1. Latent volatility	Low / high: filtered $P(\text{high}) \geq 0.50$	Two interpretable states
2. Equity market	Bear / stable / bull: Nifty 60-session return; cuts $\pm 5\%$	Separate trend and range
3. Monetary policy	Easing / hold / tightening: latest published RBI MPC cut, unchanged decision or hike	Include unchanged decisions
4. Inflation momentum	Falling / stable / rising: CPI YoY release change; cuts ± 0.25 pp	Ignore small release changes
5. Economic growth	Contraction / stable / expansion: IIP YoY first release; cuts ± 1 pp	Allow neutral growth

Timing rule	Value	Design rationale
CPI/IIP maximum age	75 calendar days	Expire stale releases
Label to position	Next native observation	Use prior information

Macro labels become usable the next Indian session after publication; exposure follows the next native date. Thresholds are fixed; CPI/IIP candidate grids were not searched.

Regime definitions and thresholds: 6–10

Dimension	States and classification	Design rationale
6. INR pressure	Appreciation / range / depreciation: prior-date USD/INR, 20-session return, cuts $\pm 1\%$	Measure medium-term currency pressure
7. Commodity cycle	Falling / range / rising: 60-session agriculture/metals/energy factor; training $q_{.33}, q_{.67}$; minimum 252 training observations	Balance categories; exclude inadequate history
8. Rates/liquidity	Loose / neutral / tight: overnight call minus repo; training cuts $-0.82, -0.74$ pp ($n = 376$); max age 7 days	Compare funding conditions with the policy rate
9. Risk sentiment	Risk on / neutral / risk off: VIX versus prior 252-session $q_{.25}, q_{.75}$; require 252 observations	Adapt thresholds to the prior annual distribution
10. Network leadership	Stock led / diffuse / hub led: 252 consecutive levels; update every 20 sessions; lag/HAC 5; raw $p < .05$; at least 3 links; balance cuts ± 0.20	Require persistent, sufficiently populated direction counts

Commodity factor: equal category and within-category weights, standardized using training SD. Leadership balance $= (\text{out} - \text{in})/(\text{out} + \text{in})$; the 2021 selection cutoff restricts labels to 2022 onward. Commodity and leadership are not estimable in the test sample under these definitions.

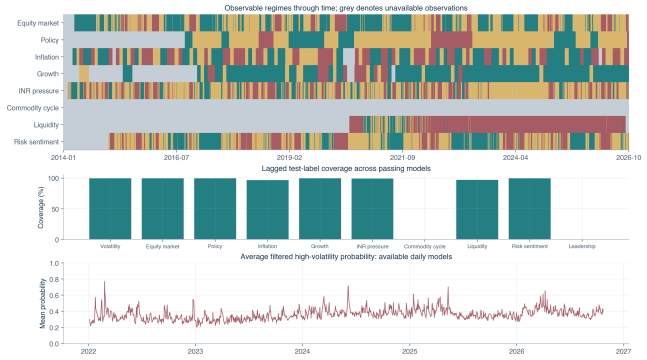
Macroeconomic context and data coverage



Series	CPI	WPI	IIP	Repo	PMI	Unemployment	Credit spread
Available months	151	153	152	154	92	18	9

Monthly snapshot: 154 reference periods and 675 index/base records. Charts exclude incomplete October 2026; dotted lines mark base changes. Snapshot values are descriptive because first-release timestamps are absent. Historical signals retain dated release records.

Regime availability and volatility through time

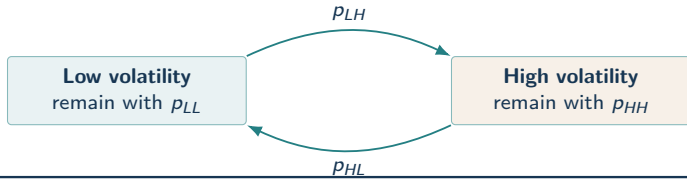


Vol.	Market	Policy	CPI	IIP	FX	Comm.	Liquidity	Risk	Leader.
99.4%	99.4%	99.4%	96.6%	99.4%	99.1%	0.0%	96.9%	99.4%	0.0%

Denominator: 290,210 predictive test observations across 1,672 passing models. Colors in the timeline correspond to the ordered states in the definition tables. The daily probability average uses the available cross-section of models.

Two-state MS-GARCH: model and interpretation

$$h_{s,t} = \omega_s + \alpha_s \varepsilon_{t-1}^2 + \beta_s \bar{h}_{t-1}, \qquad \bar{h}_{t-1} = q_{L,t-1} h_{L,t-1} + q_{H,t-1} h_{H,t-1}.$$



Element	Specification and interpretation
Input	Training-standardized stock-equation innovations; model variance after the mean dynamics
Observation density	Zero-mean Gaussian mixture; two parsimonious variance states
State ordering	$\omega_s/(1-\alpha_s-\beta_s)$ proxy; resolve state labels by relative variance
Forecast timing	Prior probabilities predict returns; the current innovation updates the filtered state

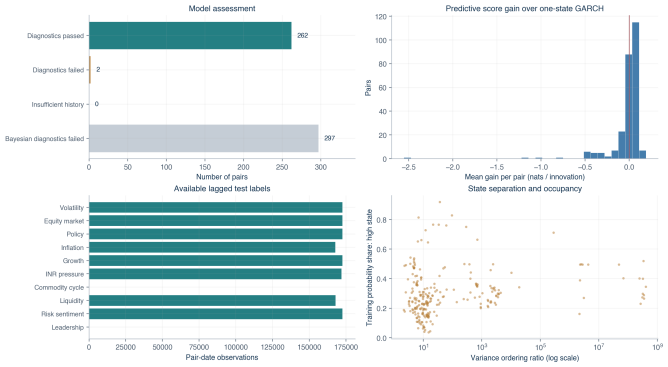
The ordering proxy is not the exact unconditional switching variance. Posterior-mean innovations do not propagate full VECM parameter uncertainty.

MS-GARCH estimation and diagnostic criteria

Choice	Setting	Design rationale
Optimizer	L-BFGS-B; 300 iterations/start	Stable bounded estimation
Initial MS coordinates	$(-3, -.5, -2, -1, 1, 2, 2, 2)$ $(-2, -1, -1, -2, 2, 1, 3, 3)$	Check two initializations
Variance intercept	$e^{-15} \leq \omega_s \leq e^{6}$ (numerical transform)	Positive finite variances
ARCH / persistence	$\alpha_s < 0.35$; $\alpha_s + \beta_s < 0.995$	Constrain shock persistence
State transitions	$0.60 < p_{LL}, p_{HH} < 0.999$	Persistent but finite states
Min. innovations: D/W/M	100 / 60 / 24	Minimum fitting history
Diagnostic gates	Convergence; ADF $p < .05$; occupancy 3–97%; ratio ≥ 1.05	Reject degenerate solutions
One-state starts	$(-2, -2, 2)$ and $(-1, -1, 1)$	Match benchmark estimation
Gaps / holdout	Reset on gaps; restart holdout in 2022	Avoid unobserved lag links
Score uncertainty	1,000 date blocks; seed 251026; lengths 20/4/2 (D/W/M)	Account for dated dependence

All settings are fixed baselines; no validation/test hyperparameter search. Numerical logistic coordinates are clipped to $[-30, 30]$. A model diagnostic pass is separate from evidence that two well-identified economic regimes exist.

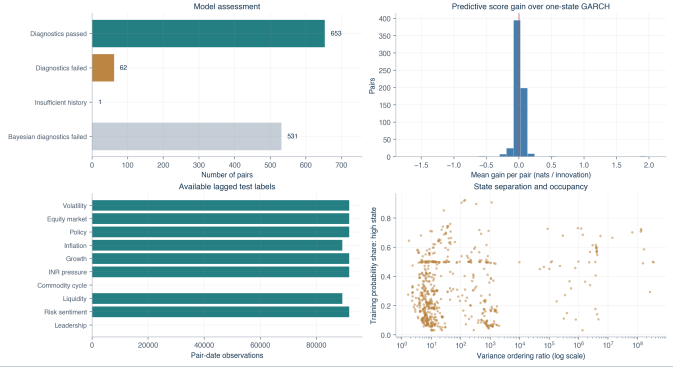
Daily horizon: model assessment



Selected pairs	Bayesian fail	MS pass	MS fail	History error	Test obs.
561	297	262	2	0	173,953

Among passing models: 206/262 have an ARCH coefficient near zero; 147/262 have high-state persistence near its lower bound. Weak identification remains material.

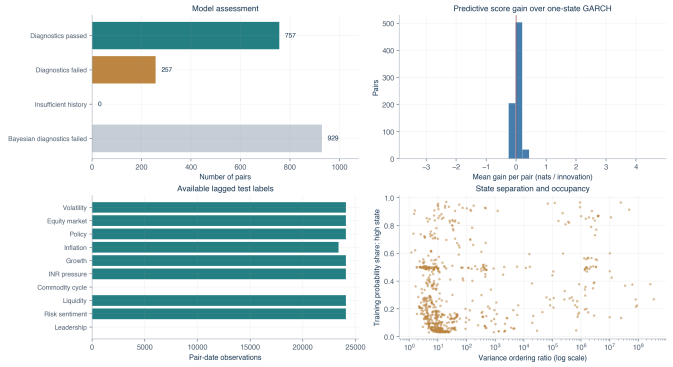
Weekly horizon: model assessment



Selected pairs	Bayesian fail	MS pass	MS fail	History error	Test obs.
1,247	531	653	62	1	92,100

Among passing models: 574/653 have an ARCH coefficient near zero; 316/653 have high-state persistence near its lower bound. Weak identification remains material.

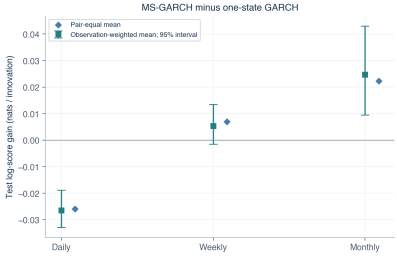
Monthly horizon: model assessment

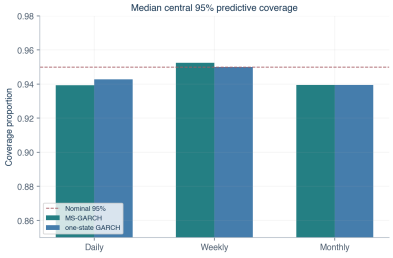


Selected pairs	Bayesian fail	MS pass	MS fail	History error	Test obs.
1,943	929	757	257	0	24,157

Among passing models: 675/757 have an ARCH coefficient near zero; 507/757 have high-state persistence near its lower bound. Weak identification remains material.

Predictive performance and forecast calibration





Horizon	Pair mean Δ	Obs. mean Δ	95% block interval	MS coverage	GARCH coverage
Daily	-0.0259	-0.0264	[-0.0329, -0.0188]	93.93%	94.28%
Weekly	+0.0070	+0.0053	[-0.0015, +0.0135]	95.24%	95.00%
Monthly	+0.0223	+0.0247	[+0.0094, +0.0431]	93.94%	93.94%

Score differences are MS-GARCH minus fitted one-state GARCH, in nats per innovation; higher is better. Intervals apply to the observation-weighted mean, conditional on selected models. Coverage is the median pair-level central 95% predictive coverage. Evidence is negative daily, inconclusive weekly, and positive monthly under observation weighting.

Synthetic hedge design and assumptions

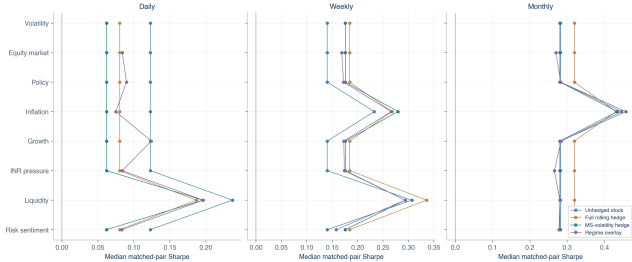
$$r_t^{\text{hedged}} = r_t^{\text{stock}} - B_{t-1} r_t^{\text{hub}} - c|\Delta B_t|, \qquad B_{t-1} = \widehat{\beta}_{t-1} P(S_{t-1} = H).$$

Choice	Value	Design rationale
Beta estimation	60 / 26 / 12 (D/W/M); clip $[-3, 3]$	Bound recent exposure
Annualization / RF	252 / 52 / 12; zero risk-free rate	Native-horizon comparability
Proxy costs	5 bp per unit hedge change, including entry/exit	Common cost proxy
Minimum observations	30 / 12 / 6 (D/W/M)	Minimum metric history
Signal timing	Prior beta / probability / regime; next return	Prevent current-return signals
Gaps / insolvency	Flat restart; costs; zero wealth after return $\leq -100\%$	Respect missing observations

Regime condition	Favorable	Neutral	Adverse
Hedge multiplier	0.25	0.50	1.00

Observable multipliers are fixed illustrations; the full state map is in the parameter companion. Returns omit currency conversion, executable rolls, financing, dividends and borrow costs.

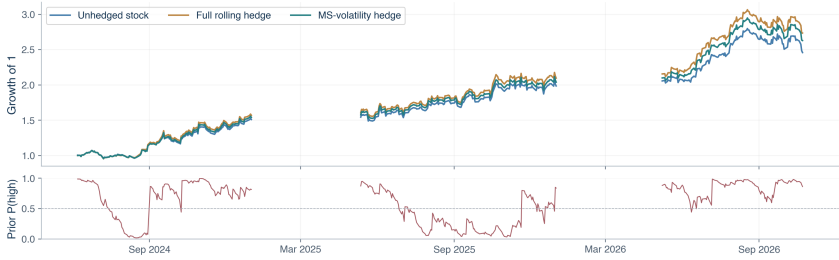
Hedge comparisons across available regimes



Horizon	Price-like pairs	Unhedged Sharpe	Full hedge Sharpe	MS-volatility Sharpe
Daily	226	0.123	0.080	0.062
Weekly	545	0.140	0.184	0.176
Monthly	483	0.282	0.319	0.280

Table: median pair Sharpe on matched volatility-rule dates. Graph: each dimension uses its own matched dates, so cross-dimension ranks are not directly comparable. All 1,254 price-like pairs are included; four full-hedge paths become insolvent. Full per-regime strategy tables accompany the slides.

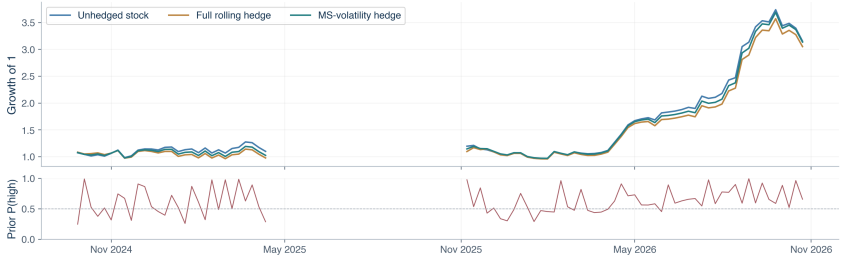
Daily illustration: Radico Khaitan and gold



Strategy	Obs.	Sharpe	Growth of 1	Ann. growth	Max. drawdown
Unhedged stock	424	2.061	2.458	70.7%	-12.2%
Full rolling hedge	424	2.269	2.735	81.8%	-11.0%
MS-volatility hedge	424	2.200	2.629	77.6%	-11.0%

RADC.NS / XAU=; 2024-06-06 to 2026-10-23. Selected by the highest MS-volatility test Sharpe among 190 eligible daily pairs, with no known source break or insolvency and at least 126 observations. Selection is retrospective; annual growth uses measured intervals and links observed segments.

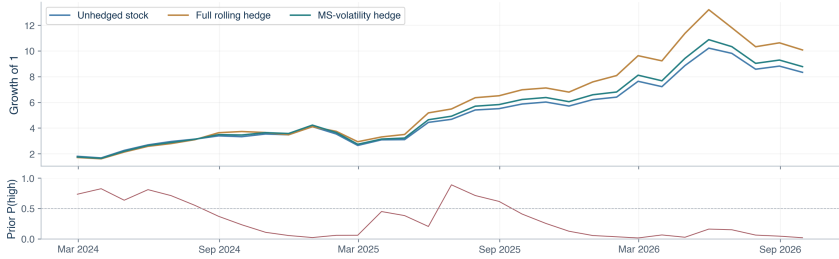
Weekly illustration: Welspun Corp and thermal coal



Strategy	Obs.	Sharpe	Growth of 1	Ann. growth	Max. drawdown
Unhedged stock	82	1.830	3.154	107.2%	-24.6%
Full rolling hedge	82	1.756	3.052	102.9%	-18.0%
MS-volatility hedge	82	1.812	3.136	106.4%	-18.7%

WGSR.NS / CXNTCLC55ECIND; 2024-09-20 to 2026-10-23. Selected by the highest MS-volatility test Sharpe among 520 eligible weekly pairs, with no known source break or insolvency and at least 52 observations. Selection is retrospective; annual growth uses measured intervals and links observed segments.

Monthly illustration: GE Vernova T&D and the INR/SGD cross



Strategy	Obs.	Sharpe	Growth of 1	Ann. growth	Max. drawdown
Unhedged stock	33	1.747	8.346	116.3%	-35.5%
Full rolling hedge	33	1.912	10.088	131.7%	-28.3%
MS-volatility hedge	33	1.809	8.791	120.4%	-35.1%

GETD.NS / INRSGD=R; 2024-01-31 to 2026-09-30. Selected by the highest MS-volatility test Sharpe among 455 eligible monthly pairs, with no known source break or insolvency and at least 24 observations. Selection is retrospective; annual growth uses measured intervals and links observed segments.

Conclusions and research limitations

Finding	Evidence	Interpretation / qualification
Directional relationships	1,034 raw-p; 178 BH links	Exploratory predictive links
Bayesian mean dynamics	1,994 sampler passes; 246 support cases	Convergence is not cointegration
Volatility modelling	1,672 MS passes; 290,210 test innovations	Regime identification is often weak
Predictive comparison	Daily -0.026; weekly $+0.007$; monthly $+0.022$ pair-mean gain	No uniform predictive advantage
Regime coverage	8 testable dimensions; 2 without valid labels	No joint ten-dimension test
Wealth illustrations	3 horizon-specific examples	Historical, selected illustrations

Research record

Complete parameter tables, pair estimates and the interactive result atlas accompany these slides. The record retains all 3,751 pair-frequency models and 30,008 directional diagnostics.

Broke Street Bets Mentor: Prof. TV Ramanathan